Lesson 2.8 - How to Make a Histogram Using a Frequency Distribution Table

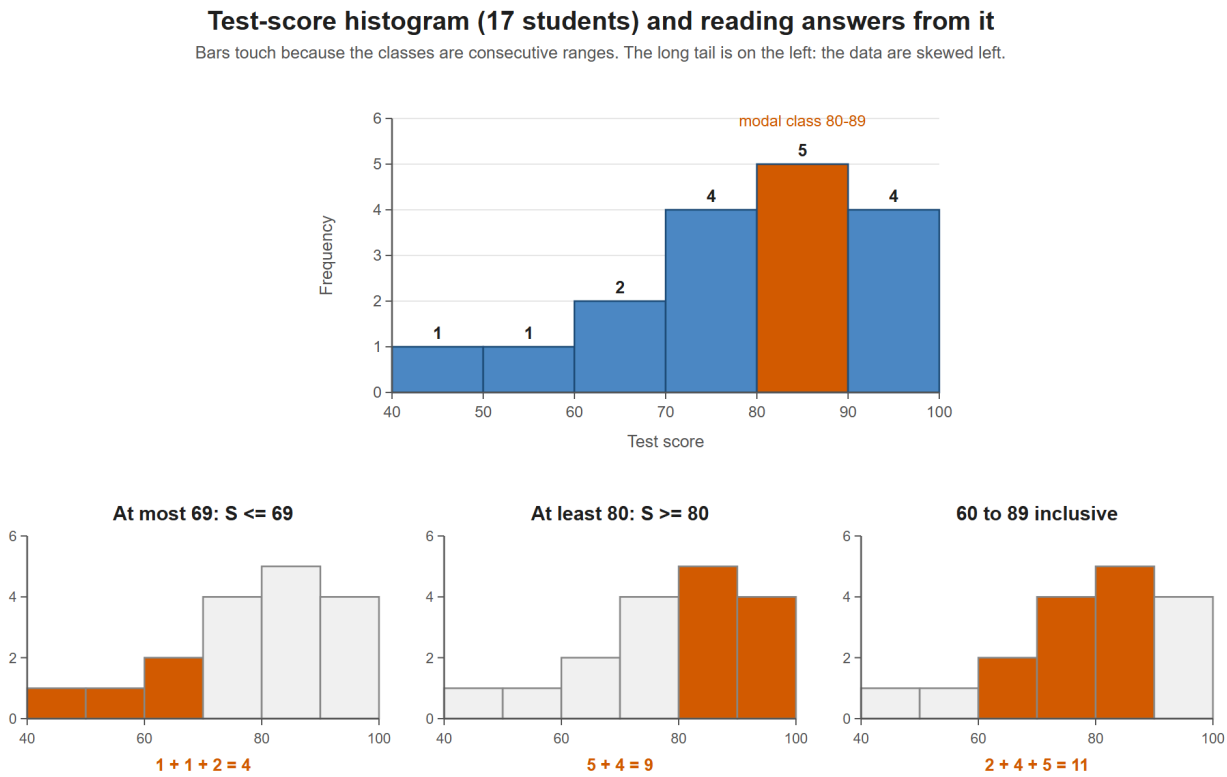
> **Video:** How To Make a Histogram Using a Frequency Distribution Table by The Organic Chemistry Tutor
> **Chapter:** 2 - Graphs and Tables

Steps

1. **Group the data into classes**, ranges of equal width. For test scores, a width of 10 is natural: 40-49, 50-59, ...
2. **Count the frequency** of each class to make a **frequency distribution table**.
3. **Draw the axes:**
   - x-axis (independent variable): the classes, such as the scores.
   - y-axis (dependent variable): the frequency.
4. **Draw a bar for each class**, with height equal to its frequency. The bars **touch**, which is what makes it a histogram rather than a bar graph.

Example - test scores of a class

Score range	Grade	Frequency
40-49	F	1
50-59	F	1
60-69	D	2
70-79	C	4
80-89	B	5
90-100	A	4
Total		**17**



The top chart is the histogram. Its bars touch, and the modal class 80-89 is shown in orange. The three small charts below shade the bars used to answer each question in the next section.

Shape and mode

- The bars are short on the left and tall on the right, so the **long tail is on the left**. The data is **skewed left** (negative skew).
- The **modal class** (the class with the highest frequency) is **80-89**.

Answering questions from a histogram

Let S be a student's score.

1. How many students scored *at most* 69? "At most 69" means $S \leq 69$, so 69 is the maximum.

$$\#(S \leq 69) = \text{sum of the frequencies of every class up to } 69 = 1 + 1 + 2 = 4 \text{ students}$$

2. How many scored *at least* 80? "At least 80" means $S \geq 80$, so 80 is the minimum.

$$\#(S \geq 80) = \text{sum of the frequencies of every class from } 80 \text{ up} = 5 + 4 = 9 \text{ students}$$

3. How many scored between 60 and 89, inclusive? "Inclusive" means 60 and 89 are counted.

$$\#(60 \leq S \leq 89) = \text{sum of the frequencies of the classes from } 60 \text{ to } 89 = 2 + 4 + 5 = 11 \text{ students}$$

> "Between 60 and 89" **exclusive** would mean 61 to 88. Check whether a question says inclusive or exclusive.

Key takeaways

- Build the **frequency distribution table first**, then draw the histogram.
- Histogram bars **touch**; bar-graph bars don't.
- Long tail on the left → skewed left. Long tail on the right → skewed right.
- "At most" = $\leq$, "at least" = $\geq$, "inclusive" = endpoints included.